

The provenance DAG as foundational knowledge architecture: a genuine gap

****No existing system — academic or production — fully implements an architecture where an immutable, append-only provenance DAG serves as the primary data structure for a knowledge system.**** This represents a real and well-documented research gap, not a solved problem repackaged. The individual components (content-addressable storage, provenance graphs, semantic layers, append-only architectures) each have mature prior art, but the specific architectural inversion — making the provenance graph *the* system rather than metadata *about* the system — has not been proposed as an integrated design in the literature. The closest existing systems reach roughly 60–80% overlap with the Atlas architecture, each missing critical elements. Multiple recent papers (2023–2026) explicitly identify the gap from different angles: knowledge graph provenance limitations, LLM knowledge decay, reproducibility crises, and AI agent memory architecture.

The architectural inversion nobody has made

The core novelty of the Atlas concept is not any single component but a **structural inversion**: in every existing system surveyed, the knowledge graph is primary and provenance is secondary metadata attached to it. Atlas proposes the reverse — the provenance DAG is the foundational structure, and the semantic knowledge graph is a derived view with every node and edge traceable through the DAG to source artifacts.

This inversion has been *gestured at* from multiple directions. **Serkan Takan (PeerJ, 2023)** directly states that "although the issue of immutability in data structures has been frequently studied, **there is no research on immutability in knowledge graphs**." [PubMed Central](https://pmc.ncbi.nlm.nih.gov/articles/PMC10495951/) Henrik Dibowski (FOIS 2024, Bosch Research) notes that "a problem that has not yet adequately been solved for KGs is the traceability and provenance of changes... KGs typically contain the current snapshot of data valid at a certain moment in time only." [Universiteit Twente] (https://www.utwente.nl/en/eemcs/fois2024/resources/papers/dibowski-full-traceability-and-provenance-for-knowledge-graphs.pdf) The comprehensive survey by **Sikos and Seneviratne (2020)** found that RDF — the dominant knowledge representation standard — "inherently lacks the mechanism to attach provenance data," [Springer](https://link.springer.com/article/10.1007/s41019-020-00118-0) and reviewed every workaround (named graphs, reification, RDF-star, singleton properties, nanopublications) without finding any fully satisfactory. [Springer](https://link.springer.com/article/10.1007/s41019-020-00118-0) [ResearchGate] (https://www.researchgate.net/publication/341258798_Provenance-Aware_Knowledge_Representation_A_Survey_of_Data_Models_and_Contextualized_Knowledge_Graphs) All of these papers identify pieces of the problem. None propose provenance-as-foundation as the solution.

The most explicit articulation of the problem Atlas solves comes from outside academia. **Jessica Talisman's "Where Provenance Ends, Knowledge Decays" (February 2026)** traces provenance as a knowledge management principle [Substack](https://jessicatalisman.substack.com/p/where-provenance-ends-knowledge-decays) from the 1841 archival concept of *respect des fonds* through the Semantic Web to the present LLM era, arguing that "LLMs strip provenance from knowledge — systematically, architecturally, and by design." [Substack](https://jessicatalisman.substack.com/p/where-provenance-ends-knowledge-decays) She notes that RAG addresses retrieval-level provenance while "leaving the deeper layer entirely unattributed." [Substack] (https://jessicatalisman.substack.com/p/where-provenance-ends-knowledge-decays) This essay calls for provenance-preserving architectures but does not propose a specific technical design.

The five closest existing systems

Each of the systems below implements a substantial subset of the Atlas architecture. The gap analysis reveals a consistent pattern: systems strong on immutability lack semantic graph layers, and systems strong on knowledge representation lack immutable provenance foundations.

TerminusDB is the closest overall match at roughly 75–80% architectural overlap. This open-source graph database (Gavin Mendel-Gleason et al., originating from Trinity College Dublin's Horizon 2020 ALIGNED project) [owlapps](https://www.owlapps.net/owlapps_apps/articles?id=62518551&lang=en) provides Git-like semantics [Wikipedia](https://en.wikipedia.org/wiki/TerminusDB) — branch, merge, time-travel — over an RDF knowledge graph [GitHub](https://github.com/terminusdb/terminusdb) using append-only succinct data structures with

delta encoding. [Grokopedia](https://grokopedia.com/page/TerminusDB) [Terminusdb](https://terminusdb.org/docs/terminusdb-explanation/) Every transaction creates a new immutable layer. [Grokopedia](https://grokopedia.com/page/TerminusDB) [Terminusdb](https://terminusdb.org/docs/terminusdb-explanation/) However, TerminusDB's foundational structure is a *versioned RDF graph*, not a provenance DAG. It tracks **what changed** but not **why it changed** or how knowledge was derived. There is no separate content-addressable blob store for raw artifacts, and no concept of transformation workers or AI processors as first-class participants.

Quit Store (AKSW, University of Leipzig) achieves roughly 60–65% overlap by providing a SPARQL 1.1 endpoint [ScienceDirect](https://www.sciencedirect.com/science/article/abs/pii/S1570826818300416) backed entirely by Git. RDF named graphs are stored in canonicalized N-Quads in Git's SHA-1 content-addressed object store, with automatic W3C PROV-O provenance generation from commit metadata. [ScienceDirect](https://www.sciencedirect.com/science/article/abs/pii/S1570826818300416) It supports `quit blame` for per-statement provenance. [CEUR-WS.org](https://ceur-ws.org/Vol-1824/mepdaw_paper_2.pdf) This is architecturally fascinating — it literally uses Git's Merkle DAG as the storage layer for a knowledge graph with provenance. What's missing: it handles only structured RDF triples (not raw artifacts like PDFs), it's an academic prototype with noted performance limitations, and the provenance is derived from version control metadata rather than being an explicit derivation DAG.

Graphiti/Zep (open-source, 2024–2025) reaches roughly 55–60% overlap and is the most relevant recent development for AI use cases. It builds temporal, provenance-aware knowledge graphs for AI agent memory [GitHub](https://github.com/getzep/graphiti) using **Neo4j** or **FalkorDB** [GitHub](https://github.com/getzep/graphiti) — the same graph databases Atlas specifies. Every entity and relationship traces back to source "episodes" (raw data ingestion events) [GitHub](https://github.com/getzep/graphiti) with bidirectional indices. Facts carry temporal validity windows (`t\_valid`, `t\_invalid`), and old facts are "invalidated, not deleted." [GitHub +3](https://github.com/getzep/graphiti) This is append-only *in spirit*, but Graphiti performs destructive entity summary updates, lacks content-addressable storage, embeds provenance in the graph rather than maintaining a separate provenance DAG, and is scoped specifically to AI agent memory rather than general knowledge management.

Fluree (founded 2017) combines an append-only immutable ledger with a semantic graph database [Crunchbase](https://www.cbinsights.com/company/fluree) supporting RDF, JSON-LD, SPARQL, and SHACL validation. Every update is cryptographically chained, enabling time-travel queries and verifiable data history. [Fluree](https://flur.ee/fluree-blog/immutable-database-explained/) It is perhaps the closest *production database* to the Atlas vision. But Fluree's immutability operates at the transactional ledger level, not at the level of a provenance DAG tracking derivation chains. There is no separate content-addressable blob store, and AI/ML processors are not modeled as first-class graph participants.

Nanopublications (Kuhn & Dumontier, ESWC 2014; ongoing) are the closest *conceptual model* from information science. Each nanopublication is an immutable, content-addressable (via "Trusty URIs" using cryptographic hashes), provenance-rich scholarly assertion containing three named RDF graphs: assertion, provenance, and publication info. Over **10 million nanopublications** exist, primarily in life sciences. Recent work (2025, International Journal of Digital Libraries) extends the model with "knowledge provenance" for multi-source assertions. The key limitation: nanopublications are a flat collection of independent atomic assertions — they don't form a provenance DAG connecting assertions through chains of derivation, and they don't support a separate semantic graph layer.

Historical and conceptual predecessors that anticipated this architecture

The deepest historical precedent is **Ted Nelson's Project Xanadu** (1960–present). Xanadu specified an immutable, add-only content space where documents are lists of pointers to regions in an "ever-growing" store, transclusion maintains "visible provenance to the source," and every connection is bidirectional. [Grokopedia](https://grokopedia.com/page/Project_Xanadu) Xanadu's "xanalogical structure" is arguably the direct ancestor of what Atlas proposes — append-only content-addressable storage with provenance as the organizing principle. The cautionary lesson: Xanadu's refusal to compromise on its complete vision prevented adoption while the simpler World Wide Web prevailed. [WebProNews](https://www.webpronews.com/project-xanadu-ted-nelsons-1960-hypertext-vision-and-enduring-legacy/)

****Pat Helland's "Immutability Changes Everything" (CIDR 2015)**** provides the theoretical and economic foundation. His core argument — "accountants don't use erasers" [Cidrdb] (https://www.cidrdb.org/cidr2015/Papers/CIDR15_Paper16.pdf) and "the truth is the log; the database is a cache of a subset of the log" [Cidrdb](https://www.cidrdb.org/cidr2015/Papers/CIDR15_Paper16.pdf) — maps directly onto Atlas's philosophy. [DEV Community](<https://dev.to/mariosangiorgio/immutability-changes-everything---pat-helland>) Helland demonstrated that cheap storage makes keeping all historical versions financially feasible, while coordination costs make immutability architecturally attractive. [Cidrdb] (https://www.cidrdb.org/cidr2015/Papers/CIDR15_Paper16.pdf) [ACM Queue] (<https://queue.acm.org/detail.cfm?id=2884038>) ****No subsequent work has explicitly applied Helland's thesis to knowledge graphs****, despite its enormous influence on event sourcing and distributed systems.

****Datomic**** (Rich Hickey, 2012–present) operationalizes Helland's thesis as a database. Every fact is an immutable "datom" (entity-attribute-value-transaction); the database is a value that only grows by accumulating new datoms; time-travel queries are native. Nubank engineers explicitly used the phrase "immutable knowledge databases" when applying Datomic to microservice dependency graphs. Datomic captures the **temporal** dimension of knowledge (what changed, when) but not the **epistemic** dimension (how knowledge was derived, from what evidence, by what process).

From archival science, the ****Records in Contexts (RiC)**** standard (ICA, v1.0 November 2023, RiC-O v1.1 May 2025) represents the most significant convergence of archival provenance theory with computational graph structures. RiC explicitly describes the archival world as "a graph of interconnected things" and models provenance as a first-class relation type (`rico:ProvenanceRelation`) expressed as an OWL ontology for RDF linked data. This is the archival profession's 180-year-old principle of provenance (**respect des fonds**) rendered as a knowledge graph standard — a conceptual ancestor of Atlas from a completely different intellectual tradition.

Adjacent architectural concepts and their relationship to the provenance DAG

****Event sourcing**** is the closest mainstream software pattern. The ****Telicent CORE platform**** implements an event-driven knowledge graph using Apache Kafka as the event log backbone, with data events flowing through topics into RDF format and multiple derived stores (graph, search index, vector store). [Telicent] (<https://telicent.io/news/event-driven-knowledge-graphs/>) The ****TMForum "Atomic Events" model**** defines append-only EAV events with timestamps for building temporal knowledge graphs. [TMForum - Inform] (<https://inform.tmforum.org/features-and-opinion/how-knowledge-graphs-based-on-atomic-events-provide-continuous-decision-intelligence>) The critical difference: event sourcing stores a linear sequence of events per aggregate, while a provenance DAG captures a richer graph of derivation relationships. No published work frames event sourcing specifically as a **knowledge management** pattern.

****Temporal knowledge graphs**** share the concern for preserving history. ****OSTRICH**** (Taelman et al., Journal of Web Semantics, 2018) provides a versioned RDF triple store with append-only delta ingestion supporting three query types across versions. [ScienceDirect] (<https://www.sciencedirect.com/science/article/abs/pii/S1570826818300404>) ****BiTRDF**** (2025) adds formal bitemporal semantics to RDF. These systems capture the **when** dimension but typically lack the **why** dimension — they don't track derivation chains or evidence.

A deep and underexplored connection exists between ****CRDTs and provenance DAGs****. Shapiro et al.'s foundational CRDT work (2011) formally proved that [Springer](https://link.springer.com/chapter/10.1007/978-3-642-24550-3_29) an ****add-only monotonic DAG is a CRDT**** — meaning an append-only provenance DAG can be replicated across distributed nodes without coordination and always merged into a consistent state. Byzantine Fault Tolerant CRDTs use Merkle-DAGs (hash graphs representing causal partial order among updates) that are structurally identical to content-addressed provenance graphs. [Jzhao](<https://jzhao.xyz/thoughts/CRDT>) This connection is ****entirely unexploited**** in the knowledge management literature — a CRDT-based provenance DAG would enable truly decentralized, coordination-free knowledge management with automatic merge, which may be the most significant unexplored implication of the Atlas architecture.

****Blue Brain Nexus**** (EPFL, Semantic Web Journal, 2023) deserves special mention as the closest working **knowledge platform** with provenance aspirations. This open-source system for neuroscience data management uses W3C PROV as its provenance backbone, SHACL validation, and an event-sourced streaming architecture, explicitly treating "provenance as a first-class citizen." [ResearchGate]

The research gap is real and multi-dimensional

The gap has been identified from at least five independent angles, none of which propose the integrated solution:

- **Knowledge graph engineering**: Sikos (2020), [Springer](<https://link.springer.com/article/10.1007/s41019-020-00118-0>) Takan (2023), [PubMed Central](<https://pmc.ncbi.nlm.nih.gov/articles/PMC10495951/>) and Dibowski (2024) document that provenance in KGs is fundamentally unsolved, with current approaches limited to metadata overlays.
- **LLM/AI provenance**: The 2025 Frontiers in Computer Science survey on KG-LLM fusion identifies "unclear knowledge provenance" as a key challenge. The **PROV-AGENT framework** (Souza et al., IEEE e-Science 2025) is the first provenance framework for AI agent workflows, extending W3C PROV with agent-specific metadata, [arXiv](<https://arxiv.org/abs/2508.02866>) but operates within traditional workflow orchestration rather than proposing a provenance-first architecture.
- **Scientific reproducibility**: With **72–83%** of researchers acknowledging a reproducibility crisis, [SEC.gov](<https://www.sec.gov/files/ctf-written-input-knowledge-provenance-protocol-kpp-revolutionary-framework-decentralized-science.pdf>) the REPRODUCE-ME ontology [Biomedcentral](<https://jbiomedsem.biomedcentral.com/articles/10.1186/s13326-021-00253-1>) (2022) and the Knowledge Provenance Protocol (KPP, 2025) both propose DAG-based provenance for science, but remain domain-specific. [PubMed Central](<https://pmc.ncbi.nlm.nih.gov/articles/PMC12376154/>)
- **Enterprise AI governance**: Industry analysts identify provenance as essential for AI compliance. Amazon Bedrock AgentCore (2025) adopted append-only memory patterns (marking outdated memories as INVALID instead of deleting). [AWS](<https://aws.amazon.com/blogs/machine-learning/building-smarter-ai-agents-agentcore-long-term-memory-deep-dive/>) But these are bolt-on solutions.
- **Content addressability for AI**: An ISPE article (January 2026) directly advocates content-addressable storage for AI knowledge management, predicting "AI copilots with built-in provenance: every answer cites the exact CIDs used." [ISPE](<https://ispe.org/pharmaceutical-engineering/january-february-2026/content-addressability-future-knowledge-management>) This comes closest to the Atlas vision from an industry perspective but lacks a complete architectural proposal.

The **ICLR 2026 Workshop on Memory for LLM-Based Agentic Systems (MemAgents)** explicitly calls for research on "provenance-aware retrieval" [OpenReview](<https://openreview.net/pdf?id=U51WxL382H>) and "structured memory access control" [arXiv](<https://arxiv.org/html/2603.10062v1>) — confirming that the research community recognizes these as open problems.

What makes the Atlas combination genuinely novel

The following table maps each Atlas architectural property against the closest existing prior art and the remaining gap:

Atlas Property	Closest Existing System	What's Missing
	---	---
Content-addressable immutable blob store (SHA256)	IPFS/IPLD, Git, DefraDB	Not integrated with knowledge graph layers
Provenance DAG as primary data structure	Nothing	all systems treat provenance as secondary This is the core architectural inversion
Semantic graph with per-edge provenance to DAG	Nanopublications, RDF-star + PROV-STAR	Nanopubs are flat collections; PROV-STAR is a bolt-on
Strictly append-only, no destructive operations	Datomic, Fluree, Arweave [DoltHub](https://www.dolthub.com/blog/2022-03-21-immutable-database/)	Not combined with knowledge graph + provenance DAG
AI/LLM as just one type of graph processor	PROV-AGENT (2025) [arXiv](https://arxiv.org/html/2508.02866v1)	Tracks agent provenance within workflows, not provenance-first

| Three-layer architecture (blobs → DAG → semantic graph) | **No system combines all three** | **The unified architecture is novel** |

The bet on accumulation over destructive summarization — predicated on large context windows becoming fast and cheap — has no direct precedent in the literature. The closest articulation is Graphiti's "non-lossy" design philosophy, [GitHub](https://github.com/getzep/graphiti) [Getzep]

(https://blog.getzep.com/content/files/2025/01/ZEP__USING_KNOWLEDGE_GRAPH_TO_POWER_LLM_AGENT_MEMORY_202501170 but Graphiti still performs destructive entity summary updates. The explicit framing of "accumulate everything, context windows will catch up" as an architectural strategy appears to be original.

Conclusion

Atlas sits at a genuine intersection of multiple well-established research traditions — archival provenance theory, content-addressable data structures, immutable database architectures, temporal knowledge graphs, and W3C provenance standards — but **synthesizes them in a way that has not been previously proposed**. The architectural inversion (provenance DAG as foundation, knowledge graph as derived view) is the key novel contribution. The closest systems are TerminusDB (immutable versioned knowledge graph without derivation tracking), Quit Store (Git-backed SPARQL with PROV-O), [ResearchGate]

(https://www.researchgate.net/publication/303947118_Distributed_Collaboration_on_RDF_Datasets_Using_Git_Towards_the_Quit-Store) and Graphiti/Zep (temporal provenance-aware AI memory without immutability). [GitHub]

(https://github.com/getzep/graphiti) The most important underexplored theoretical connection is to **CRDTs**: since an add-only monotonic DAG is provably a CRDT, the Atlas architecture could natively support coordination-free distributed knowledge management — a property no existing knowledge system offers. The richest intellectual lineage runs through Xanadu (immutable add-only content with visible provenance), Helland (immutability as architectural principle), W3C PROV (formal provenance vocabulary), and RiC-O (archival provenance as graph structure). The research gap is not merely "nobody has built this yet" — it is that the specific architectural pattern of provenance-as-substrate has not been **articulated as a design principle** in the literature, even as multiple communities independently identify the problems it would solve.